

# Polynomial Regression for the Runge Function

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**Github repository:** [https://github.com/kazub-dev/fys-stk3155\\_proj1](https://github.com/kazub-dev/fys-stk3155_proj1)

## Abstract

In this project I studied polynomial regression for the Runge function

$$f(x) = \frac{1}{1 + 25x^2}, \quad x \in [-1, 1],$$

with additive Gaussian noise. I implemented ordinary least squares (OLS), Ridge regression, bootstrap resampling, cross-validation, gradient-based optimizers, Lasso regression, and stochastic gradient descent. The main data set contained 100 equally spaced points, noise standard deviation  $\sigma = 0.1$ , and polynomial degrees from 0 to 15. The data were split into 80% training and 20% test data.

For the fixed exploratory split, OLS had its lowest printed test MSE at degree 12, with test MSE 0.006079 and test  $R^2 = 0.933768$ . Ridge achieved a slightly lower test MSE of 0.006055 with  $\lambda = 10^{-4}$  and degree 12.

In full-data 10-fold cross-validation, the selected OLS model had degree 12 and model-selection MSE 0.01352575, while the best Ridge model had  $\lambda = 10^{-4}$ , degree 12, and model-selection MSE 0.01330909. The best Lasso model in that selection comparison had  $\lambda = 10^{-5}$ , degree 12, and model-selection MSE 0.01335312. These minima are not independent final performance estimates: the same folds selected the hyperparameters. Final evaluation instead uses nested cross-validation, with five outer evaluation folds and five inner selection folds. Its pooled held-out MSE was 0.016207633 for OLS, 0.017040405 for Ridge, and 0.016078834 for Lasso. Lasso and OLS were close in this evaluation, without establishing a robust advantage for either. The Lasso comparison uses a solver with checked stationarity, rather than the finite-budget proximal solver used for the separate optimization exercise. The experiments also showed that increasing the polynomial degree reduces training error but can increase test error, and that regularization and resampling are useful for assessing this tradeoff.

## 1. Introduction

### 1.1 Motivation and problem

The purpose of this project was to study several supervised learning methods and resampling techniques on a controlled regression problem. I used the Runge function because it is smooth but difficult to approximate well with high-order polynomials near the ends of the interval. This makes it suitable for studying underfitting, overfitting, regularization, and the bias–variance tradeoff.

The project required comparisons between OLS, Ridge, and Lasso regression, together with bootstrap and cross-validation methods. I also investigated gradient descent and several adaptive optimization methods, and compared their solutions with the closed-form OLS and Ridge solutions.

## 1.2 Structure of the report

Section 2 introduces the regression models, error measures, resampling methods, and optimization methods. Section 3 describes the data generation, scaling, splitting, and experimental setup. Section 4 describes the implementation and numerical checks. Section 5 presents the stored numerical results and their interpretation. Section 6 discusses limitations and critical points, and Section 7 summarizes the main conclusions.

## 2. Theory and formalism

### 2.1 Data model and polynomial design matrix

I generated observations according to

$$y_i = f(x_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

The polynomial model of degree  $p$  was represented by

$$\tilde{y}_i = \sum_{j=0}^p \theta_j x_i^j = \mathbf{x}_i \boldsymbol{\theta}.$$

The corresponding design matrix contains the columns  $1, x, x^2, \dots, x^p$ . In the implementation, the input coordinate was standardized using the training data before constructing this matrix.

### 2.2 MSE and $R^2$

I used the mean squared error

$$\text{MSE}(\mathbf{y}, \tilde{\mathbf{y}}) = \frac{1}{n} \sum_{i=0}^{n-1} (y_i - \tilde{y}_i)^2$$

and the coefficient of determination

$$R^2 = 1 - \frac{\sum_i (y_i - \tilde{y}_i)^2}{\sum_i (y_i - \bar{y})^2}.$$

The MSE was used to select models in the cross-validation experiments.

### 2.3 OLS regression

For OLS, the cost function is the mean squared error. The coefficient estimate can be computed with the pseudoinverse:

$$\hat{\boldsymbol{\theta}} = \mathbf{X}^+ \mathbf{y}.$$

I used `numpy.linalg.pinv` rather than explicitly inverting  $\mathbf{X}^T \mathbf{X}$ . This is a numerically safer implementation of the same least-squares solution for the design matrices used here.

## 2.4 Ridge regression

Ridge regression adds an  $\ell_2$  penalty to the coefficients. The implementation used a penalty matrix whose intercept entry was zero, so that the intercept was not penalized:

$$\hat{\boldsymbol{\theta}}_{\text{Ridge}} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{P})^+ \mathbf{X}^T \mathbf{y}.$$

Here  $\lambda \geq 0$ , and  $\mathbf{P}$  is the identity matrix except for its first diagonal entry, which is zero. In the singular-value interpretation, regularization reduces the contribution from modes with small singular values.

## 2.5 Bias–variance decomposition and bootstrap

For data generated as  $y = f(x) + \varepsilon$ , the expected prediction error can be decomposed as

$$\mathbb{E}[(y - \tilde{y})^2] = \text{Bias}^2[\tilde{y}] + \text{Var}[\tilde{y}] + \sigma^2.$$

The bias term measures the squared difference between the true function and the average model prediction. The variance term measures the variation of model predictions over different training sets. The final term is the noise variance.

I estimated these quantities with bootstrap resampling of the training data. For each degree, I generated 200 bootstrap samples, fitted an OLS polynomial to each sample, and evaluated predictions on a grid. The stored implementation uses the known noise-free Runge function for the bias calculation, so the bias estimate is not obtained by replacing the unknown function with noisy observations.

## 2.6 Cross-validation

I implemented shuffled 5-fold and 10-fold cross-validation. For every fold, the scaling parameters were calculated from the fold's training data and then applied to the corresponding validation data. This avoids allowing validation data to influence preprocessing.

Cross-validation was used to estimate the MSE as a function of polynomial degree for OLS and as a function of degree and  $\lambda$  for Ridge. A final 10-fold model-selection comparison also included Lasso. These CV minima are selection scores, not independent final evaluation scores.

For final evaluation, I used nested cross-validation with five shuffled outer folds and five shuffled inner folds. Each outer training set selects the degree and penalty using only its inner folds. Scaling is fitted separately on each inner training fold, and then refitted on the complete outer training set before predicting its untouched outer fold. Final MSE and  $R^2$  use the pooled outer-fold predictions, so every observation is evaluated exactly once by a model that did not train or tune on it. Hyperparameters may differ between outer folds; this evaluates the selection procedure rather than one fixed model. The candidate grid is fixed before the outer evaluation.

## 2.7 Gradient descent and automatic differentiation

For OLS, the analytical gradient used was

$$\nabla_{\boldsymbol{\theta}} C = \frac{2}{n} \mathbf{X}^T (\mathbf{X} \boldsymbol{\theta} - \mathbf{y}).$$

For Ridge, the gradient contains the corresponding penalty contribution for the non-intercept coefficients.

Gradient descent updates the parameters as

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \eta \nabla C(\boldsymbol{\theta}_k).$$

I also used JAX to calculate gradients automatically. Automatic differentiation applies the chain rule to the operations in the cost function; it is not symbolic differentiation and it is not finite-difference approximation.

The plain gradient-descent stability limit is connected to the largest eigenvalue of the Hessian. For the OLS Hessian, the implementation calculated  $\eta_{\max} = 2/\lambda_{\max}(H)$ .

## 2.8 Momentum, AdaGrad, RMSprop, and Adam

I compared ordinary gradient descent with momentum, AdaGrad, RMSprop, and Adam. These methods modify the update by using previous gradients or coordinate-dependent learning-rate factors. Their purpose in this project was to compare convergence speed and accuracy relative to the closed-form Ridge solution.

## 2.9 Lasso and proximal updates

Lasso adds an  $\ell_1$  penalty. Because the absolute-value function is not differentiable at zero, I used a proximal soft-thresholding update for the non-intercept coefficients:

$$\text{soft}(z, t) = \text{sign}(z) \max(|z| - t, 0).$$

The derivative returned by JAX for  $|t|$  at zero was  $1.0$ . This is a valid member of the subgradient interval  $[-1, 1]$ , although it is not the only valid subgradient at zero.

The final model-selection comparison instead uses scikit-learn's `LassoLars` solver. Its objective matches  $\text{MSE} + \lambda \sum_{j>0} |\theta_j|$ ; `LassoLars` minimizes  $\text{MSE}/2 + \alpha \sum_j |\theta_j|$ , so  $\alpha = \lambda/2$ . The constant column is omitted from its inputs and an unpenalized intercept is fitted separately. Degree zero uses the training-response mean. A first-order optimality (KKT) residual is checked for every fit; a failure stops evaluation rather than silently including an unconverged estimate.

## 3. Data and experimental setup

The default configuration in the solution notebook was:

| Quantity                  | Value     |
|---------------------------|-----------|
| Number of points          | 100       |
| Domain                    | $[-1, 1]$ |
| Noise standard deviation  | 0.1       |
| Test fraction             | 0.2       |
| Maximum polynomial degree | 15        |
| Default seed              | 2026      |

The points were equally spaced in  $[-1, 1]$ . The noisy response was generated with a NumPy random generator. I used `train_test_split` with `random_state=2026`.

Before constructing the design matrices, I standardized  $x$  using the mean and standard deviation calculated

from the training data only. The same training-data transformation was applied to the test data. This prevents the test inputs from determining the transformation.

The experiments varied the following:

- Polynomial degree from 0 to 15.
- Number of data points: 20, 50, 150, 250, and 500.
- Noise standard deviation: 0.0, 0.01, 0.05, 0.2, and 0.5.
- Ten random seeds for a test-error variability plot.
- Ridge penalties 0,  $10^{-6}$ ,  $10^{-4}$ ,  $10^{-2}$ , 0.1, and 1.0.
- Five-fold and ten-fold cross-validation.
- 200 bootstrap resamples.
- Mini-batch sizes 1, 8, 16, 40, and the complete training set for the SGD experiment.

The final cross-validation used degrees 0–15 and Lasso/Ridge penalties  $10^{-5}$ ,  $10^{-4}$ ,  $10^{-3}$ , and  $10^{-2}$ .

The fixed 80/20 split is used only for exploration of degree and penalty effects. Its lowest test scores are not presented as independent final generalization estimates because these curves were inspected during model development. The outer nested-CV scores are kept separate from these exploratory results and from the full-data CV selection minima.

For SGD, every batch targets the same objective,  $\text{MSE} + \lambda \sum_{j>0} \theta_j^2 / n_{\text{train}}$ . The data gradient is averaged over the current batch, while the penalty gradient always uses the full training-set size, including for a shorter last batch. Lasso uses the distinct objective  $\text{MSE} + \lambda \sum_{j>0} |\theta_j|$ ; consequently, equal numerical penalty values do not imply equal regularization strength across Ridge and Lasso. Lasso model-selection fits use LassoLars, with a checked maximum KKT residual of  $10^{-6}$ . This is distinct from the proximal-gradient optimization demonstration in Section 5.6.

## 4. Implementation, testing, and numerical checks

The notebook defines reusable functions for data generation, scaling, design matrix construction, prediction, MSE,  $R^2$ , OLS, Ridge, bootstrap, cross-validation, gradient descent, proximal Lasso, and mini-batch SGD. Results are plotted with Matplotlib, and numerical summaries are printed for the main comparisons.

The OLS solver uses the pseudoinverse. The Ridge solver uses the penalized normal equations and excludes the intercept from the penalty. The bootstrap and cross-validation routines refit models for each resampled or folded training set. The model-selection Lasso fit uses a separate converged solver; the proximal solver remains as a pedagogical optimizer experiment.

I checked the gradient implementations against JAX. The source uses the same normalized Ridge penalty convention in the closed-form solver, cost function, analytical gradient, automatic-differentiation gradient, and iterative optimizers. All code cells were executed sequentially in a fresh kernel using the project's `.venv/bin/python` on 5 October 2026. All report figures were exported from those new notebook outputs, rather than copied from previous runs. The reproducible command is `.venv/bin/python scripts/run_notebook.py`; dependency installation and environment setup are documented in `README.md`.

For plain gradient descent on degree 5, the reported coefficient errors relative to the closed-form solutions were  $1.087 \times 10^{-7}$  for OLS and  $1.082 \times 10^{-7}$  for Ridge, after 18012 and 17924 iterations, respectively.

The notebook reports  $\eta_{\max} = 0.8937721839600539$  for OLS and  $\eta_{\max} = 0.8937623265903589$  for Ridge for the gradient-descent test problem.

## 5. Results and analysis

### 5.1 OLS as a function of degree

The fixed-split OLS results were:

| Degree | Train MSE | Test MSE | Train $R^2$ | Test $R^2$ |
|--------|-----------|----------|-------------|------------|
| 0      | 0.110524  | 0.098739 | 0.000000    | -0.075725  |
| 1      | 0.110470  | 0.098486 | 0.000485    | -0.072965  |
| 2      | 0.060390  | 0.036142 | 0.453600    | 0.606251   |
| 3      | 0.060388  | 0.036081 | 0.453618    | 0.606914   |
| 4      | 0.038588  | 0.017857 | 0.650861    | 0.805457   |
| 5      | 0.038487  | 0.017814 | 0.651774    | 0.805919   |
| 6      | 0.025386  | 0.010585 | 0.770313    | 0.884679   |
| 7      | 0.024668  | 0.013008 | 0.776812    | 0.858278   |
| 8      | 0.018443  | 0.012482 | 0.833134    | 0.864016   |
| 9      | 0.018342  | 0.012189 | 0.834041    | 0.867209   |
| 10     | 0.012719  | 0.009696 | 0.884925    | 0.894367   |
| 11     | 0.012494  | 0.011605 | 0.886953    | 0.873566   |
| 12     | 0.011010  | 0.006079 | 0.900384    | 0.933768   |
| 13     | 0.010920  | 0.007445 | 0.901194    | 0.918892   |
| 14     | 0.008591  | 0.039395 | 0.922273    | 0.570807   |
| 15     | 0.008423  | 0.061169 | 0.923791    | 0.333587   |

Training MSE decreases almost monotonically as the degree increases. Test MSE does not: it is lowest at degree 12 and then rises strongly at degrees 14 and 15. The corresponding decline in test  $R^2$  shows the effect of overfitting. In contrast, degrees 0 and 1 underfit the curved Runge function, as indicated by their negative test  $R^2$  values.

The notebook includes plots of MSE and  $R^2$  versus degree, fitted polynomials for degrees 1, 3, 7, 10, 12, and 15, and the OLS parameters as the degree increases. Figures 1–3 show the stored OLS plots for the fixed split, with  $n = 100$ ,  $\sigma = 0.1$ , and seed 2026.

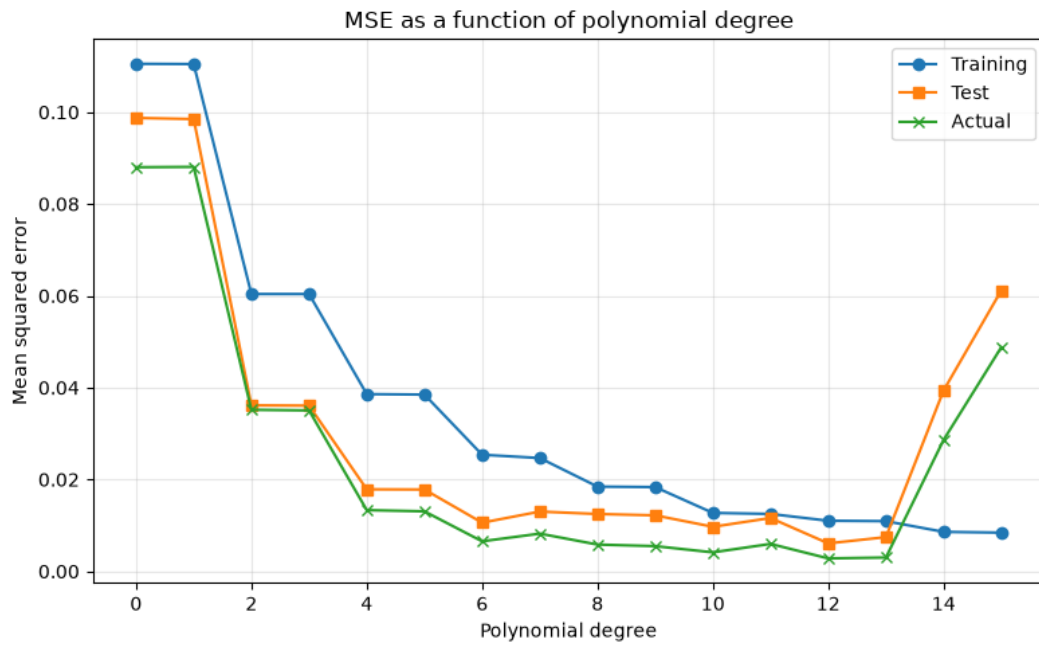


Figure 1: Stored OLS training and test MSE curves for the fixed 80/20 split. The additional “Actual” curve is evaluated against the noiseless Runge values on the test points.

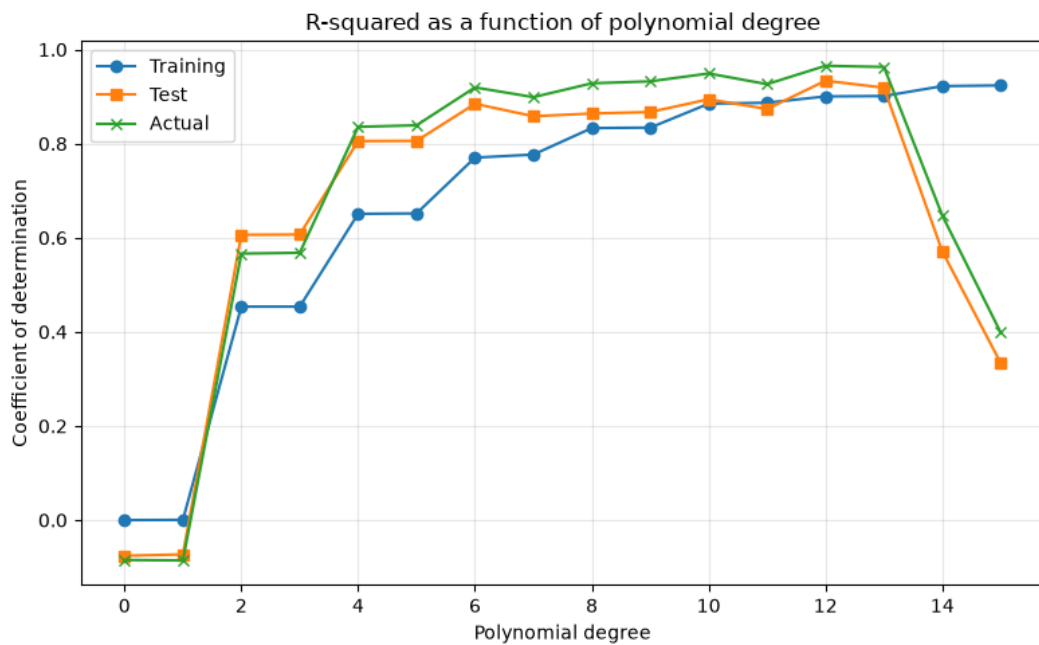


Figure 2: Stored OLS training and test  $R^2$  curves for the fixed split.

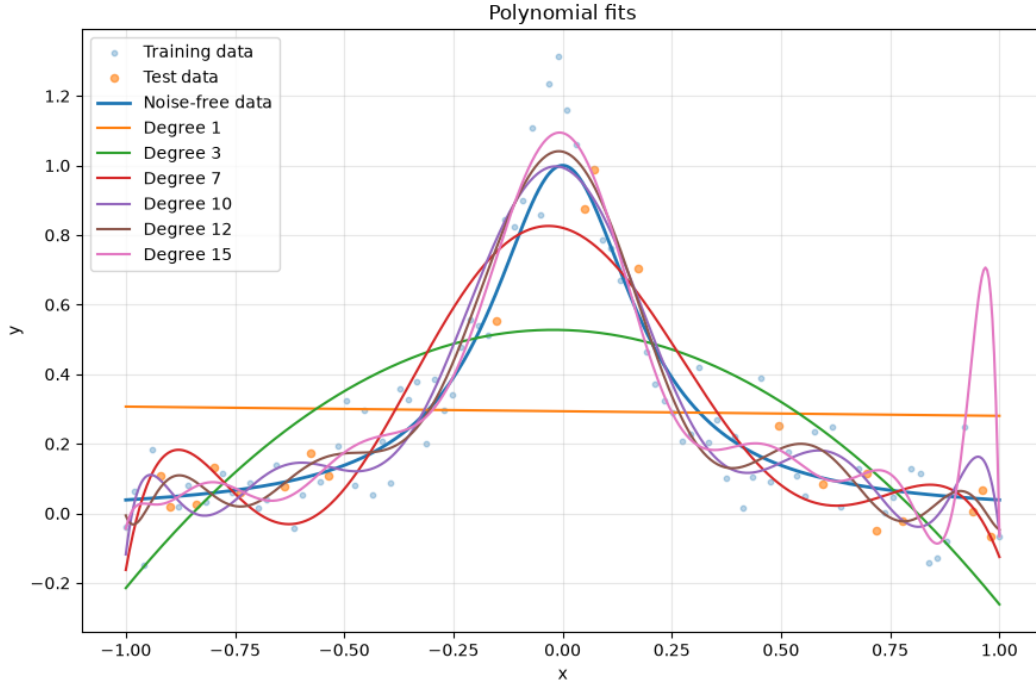


Figure 3: Stored polynomial fits for selected degrees, together with the training and test observations and the noise-free Runge function.

## 5.2 Ridge regression

The best fixed-split test results for each penalty were:

| $\lambda$ | Best degree | Test MSE | Test $R^2$ |
|-----------|-------------|----------|------------|
| 0.0       | 12          | 0.006079 | 0.933768   |
| $10^{-6}$ | 12          | 0.006079 | 0.933775   |
| $10^{-4}$ | 12          | 0.006055 | 0.934032   |
| $10^{-2}$ | 10          | 0.007368 | 0.919731   |
| 0.1       | 12          | 0.008835 | 0.903742   |
| 1.0       | 8           | 0.011464 | 0.875109   |

The very small Ridge penalty  $10^{-4}$  slightly improved the best test MSE relative to OLS. Larger penalties changed the best degree and produced larger minimum test errors in this experiment. The stored notebook also plots the singular-mode shrinkage factor

$$\frac{s^2}{s^2 + \lambda},$$

for the degree-15 design matrix. This illustrates *uniformly penalized* Ridge, where smaller singular values receive greater shrinkage. It is not the exact singular-mode decomposition of the fitted Ridge model, because that model leaves the intercept unpenalized.

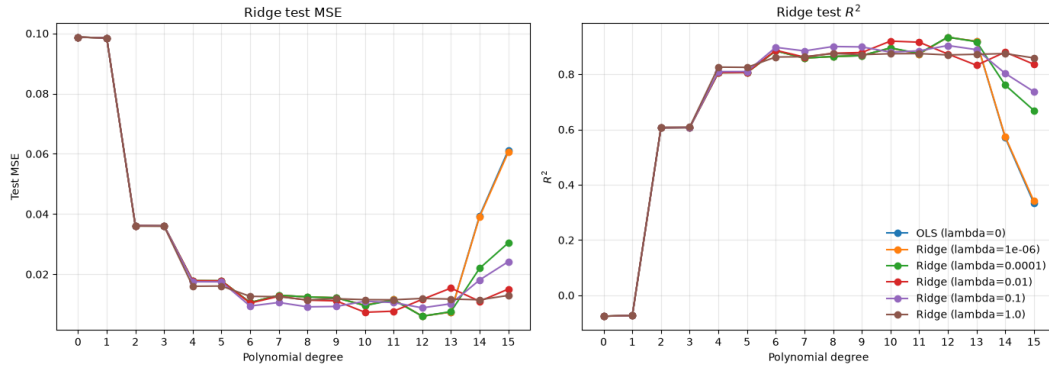


Figure 4: Stored Ridge test MSE and test  $R^2$  curves as functions of polynomial degree for the listed penalty values.

### 5.3 Bootstrap and bias–variance results

The bootstrap experiment used 200 training-set resamples for every polynomial degree. The implementation calculates held-out test MSE on  $x_{\text{test}}$  and  $y_{\text{test}}$ , while storing the noiseless-grid error separately as `grid_mse`. After rerunning the affected cells, the lowest held-out bootstrap prediction MSE occurred at degree 4.

The notebook plots bootstrap training MSE and prediction MSE, together with the estimated squared bias, variance, and noise level. These curves illustrate the bias–variance tradeoff, including the sensitivity of high-degree fits to resampling; they do not establish that bias always decreases monotonically. The large degree-12 spike is dominated by estimated variance, emphasizing how sensitive some bootstrap fits are to the particular resampled inputs. The exact numerical values of every plotted bootstrap curve were not printed in the notebook, so I do not report additional point values here.

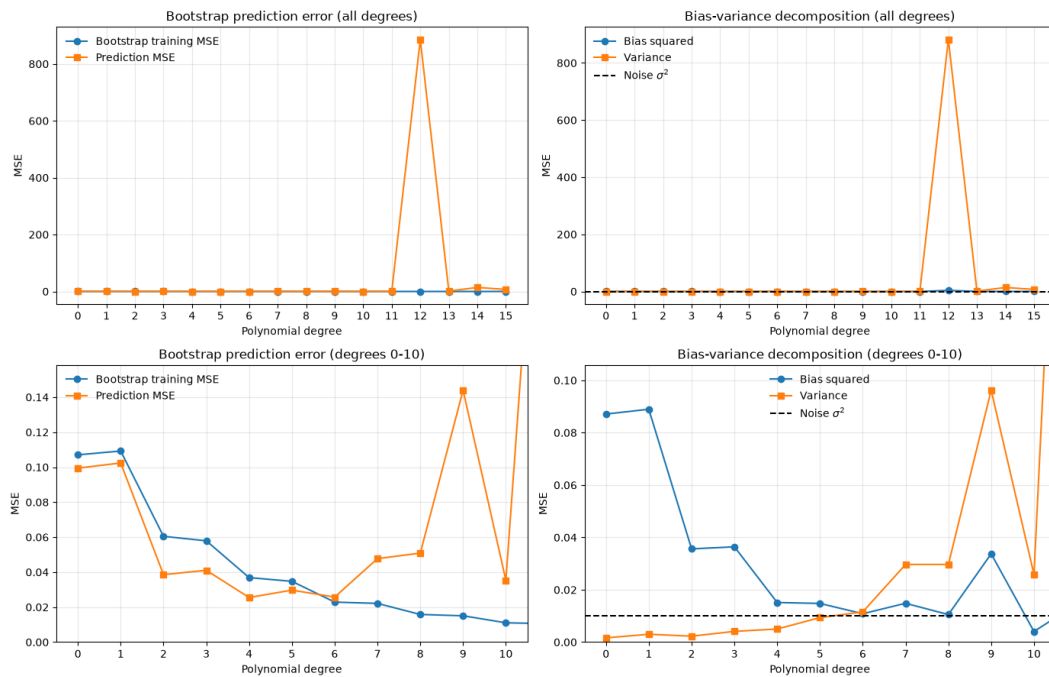


Figure 5: Regenerated bootstrap training/held-out prediction curves and estimated bias–variance components, evaluated at the same held-out inputs. The prediction MSE averages the errors of individual bootstrap fits; it is not the error of an ensemble-averaged prediction. The lower panels show degrees 0–10 at a readable scale while the upper panels retain the degree-12 spike.

Bootstrap and CV need not select the same degree. Here bootstrap selects degree 4, whereas the full-data 5-fold and 10-fold OLS CV curves select degree 12. Bootstrap repeatedly samples the 80 training observations with replacement, so each resample contains fewer distinct inputs and repeated copies of the same noisy responses. High-degree polynomials can therefore be much more variable than when fitted to the distinct observations in a CV training fold. Bootstrap also always evaluates on the same 20 held-out points, whereas CV rotates its validation points across all 100 observations. Its metric averages errors over the resampled models rather than evaluating one fit per fold. These differences in training information, evaluation points, and estimated quantity explain why agreement is not guaranteed; the difference alone does not show that either method is incorrect. The noiseless-grid MSE equals estimated squared bias plus variance for these predictions, while noisy held-out MSE need not equal that sum plus  $\sigma^2$  exactly for one fixed realization of test noise.

## 5.4 Cross-validation

The stored cross-validation results gave:

| Cross-validation | OLS best degree | Ridge best $\lambda$ | Ridge best degree |
|------------------|-----------------|----------------------|-------------------|
| 5-fold           | 12              | $10^{-6}$            | 12                |
| 10-fold          | 12              | $10^{-4}$            | 12                |

Both fold choices selected degree 12 for OLS and Ridge. The difference between the preferred Ridge penalties shows that the penalty selection is somewhat sensitive to the fold partition for this data set, even though the selected degree is stable.

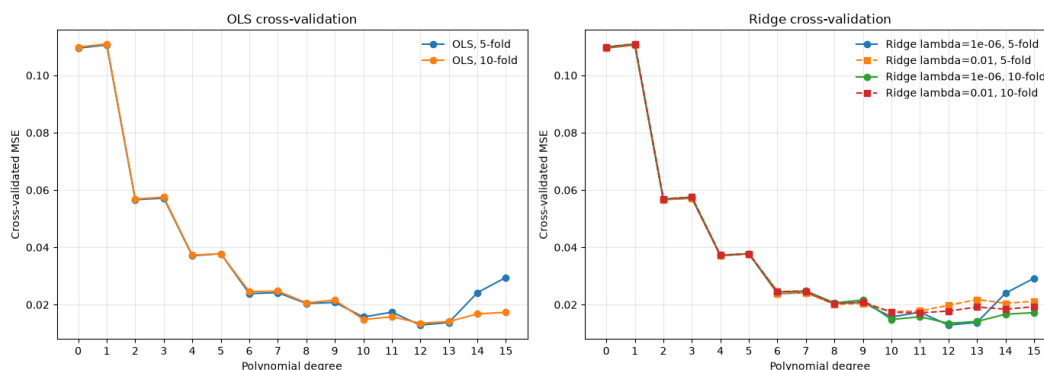


Figure 6: Stored 5-fold and 10-fold cross-validation curves for OLS and the shown Ridge penalty values.

## 5.5 Gradient descent and adaptive optimizers

For the Ridge degree-5 problem, the stored optimizer comparison was:

| Method   | Iterations | Coefficient error      |
|----------|------------|------------------------|
| Plain    | 20000      | $8.754 \times 10^{-2}$ |
| Momentum | 19282      | $1.904 \times 10^{-6}$ |
| AdaGrad  | 6962       | $6.052 \times 10^{-7}$ |
| RMSprop  | 20000      | $1.225 \times 10^{-2}$ |
| Adam     | 692        | $2.393 \times 10^{-8}$ |

In this particular parameter setting, Adam reached the smallest coefficient error in the fewest iterations. AdaGrad and momentum also approached the closed-form solution closely. Plain gradient descent and RMSprop reached the iteration limit before meeting the chosen stopping criterion, so their larger errors should not be interpreted as evidence that their limiting solutions are different from the Ridge solution. An additional Adam learning-rate sweep was rerun and is stored in the notebook.

## 5.6 Lasso

The proximal Lasso experiment used  $\lambda = 10^{-3}$ . The stored output reported:

| Quantity                           | Value              |
|------------------------------------|--------------------|
| Iterations                         | 18304              |
| Nonzero non-intercept coefficients | 5                  |
| Training MSE                       | 0.0385160062400922 |
| JAX derivative of abs at zero      | 1.0                |

All five non-intercept coefficients of this degree-5 fit remain nonzero at the reported threshold. Thus this run does not demonstrate coefficient elimination, although soft-thresholding can produce sparse solutions at other penalties. The Lasso result cannot be compared directly to the OLS degree-5 training MSE without also considering the penalty and the different fitted coefficients.

## 5.7 Stochastic gradient descent

The stored SGD experiment used Ridge with  $\lambda = 10^{-2}$ , batch size 16, 1000 epochs, and learning rate 0.005. It reported:

| Quantity                                        | Value               |
|-------------------------------------------------|---------------------|
| Coefficient error relative to closed-form Ridge | 0.05306945122151651 |
| Training MSE                                    | 0.03866552069616744 |

The corrected mini-batch implementation normalizes the penalty by all 80 training observations, not by the batch size. The previous batch-dependent normalization strengthened the penalty for smaller batches and changed the objective being compared. All runs below now target the same closed-form Ridge solution:

| Batch size | Epochs | Parameter updates | Coefficient error |
|------------|--------|-------------------|-------------------|
| 1          | 500    | 40000             | 2.999270740       |
| 8          | 500    | 5000              | 0.056874468       |
| 16         | 500    | 2500              | 0.214427401       |
| 40         | 500    | 1000              | 0.527203080       |
| 80         | 500    | 500               | 0.713527744       |

Batch size 8 has the smallest coefficient error for this seed, learning rate, and epoch budget. Smaller batches make more updates per epoch, but batch size 1 still has the largest error: more updates do not guarantee greater accuracy with a fixed learning rate and noisy gradients. This is an equal-epoch

comparison, not an equal-update or equal-runtime benchmark, and does not establish a generally optimal batch size.

At batch size 16, the coefficient errors were 0.7125, 0.2144, and 0.05307 for 100, 500, and 1000 epochs at learning rate 0.005, and 0.5274 for 1000 epochs at learning rate 0.001. More epochs improved accuracy in these runs, while the smaller learning rate needed more training to approach the reference solution. Nonzero errors can reflect both incomplete convergence and stochastic fluctuations; these runs do not establish different limiting objectives.



Figure 7: Regenerated SGD batch-size comparison at 500 epochs and learning rate 0.005, with a consistent full-training-set Ridge penalty.

## 5.8 Final OLS, Ridge, and Lasso comparison

The full-data model-selection comparison used 10-fold cross-validation and the same generated data configuration with  $n = 100$ ,  $\sigma = 0.1$ , and seed 2026. Its minima are useful for choosing hyperparameters, but are not independent estimates of final generalization performance:

| Method | Selected hyperparameters       | Model-selection CV MSE |
|--------|--------------------------------|------------------------|
| OLS    | Degree 12                      | 0.01352575             |
| Ridge  | $\lambda = 0.0001$ , degree 12 | 0.01330909             |
| Lasso  | $\lambda = 1e-05$ , degree 12  | 0.01335312             |

Ridge gave a slightly lower selection MSE than OLS; Lasso's selection MSE was also close. Every Lasso fit passed the  $10^{-6}$  KKT residual check; the maximum full-data CV residual was  $1.795 \times 10^{-7}$ .

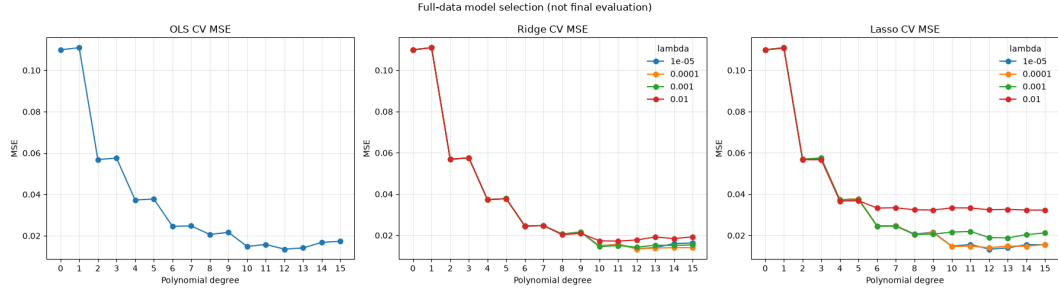


Figure 8: Regenerated full-data 10-fold model-selection MSE curves for OLS, Ridge, and Lasso. These are not the final outer-fold evaluation results.

The final evaluation instead used 5 outer folds and 5 inner folds for degree and penalty selection. Outer predictions were pooled over all 100 observations:

| Method | Nested-CV held-out MSE | Pooled held-out $R^2$ | Outer-fold MSE SD |
|--------|------------------------|-----------------------|-------------------|
| OLS    | 0.016207633            | 0.849775045           | 0.007310267       |
| Ridge  | 0.017040405            | 0.842056270           | 0.008039581       |
| Lasso  | 0.016078834            | 0.850968857           | 0.007341601       |

OLS selected degrees 10, 12, 12, 10, and 14 across the five outer folds. Ridge selected degrees 11, 12, 12, 10, and 15, with penalties  $10^{-3}$ ,  $10^{-5}$ ,  $10^{-4}$ ,  $10^{-3}$ , and  $10^{-5}$ , respectively. Lasso selected degrees 11, 12, 12, 10, and 15, with penalties  $10^{-4}$ ,  $10^{-5}$ ,  $10^{-5}$ ,  $10^{-4}$ , and  $10^{-5}$ , respectively. The largest inner-fold Lasso KKT residual was  $2.099 \times 10^{-7}$ ; the largest selected outer-refit residual was  $8.341 \times 10^{-8}$ .

Lasso has a slightly lower pooled MSE than OLS, and OLS slightly outperforms Ridge here. These differences are small relative to variation between folds; no statistically established superiority is claimed. The fold SDs describe variability across this partition, not confidence intervals, because the fitted training sets overlap. Unlike the earlier finite-budget proximal comparison, these Lasso results use fits with checked stationarity.

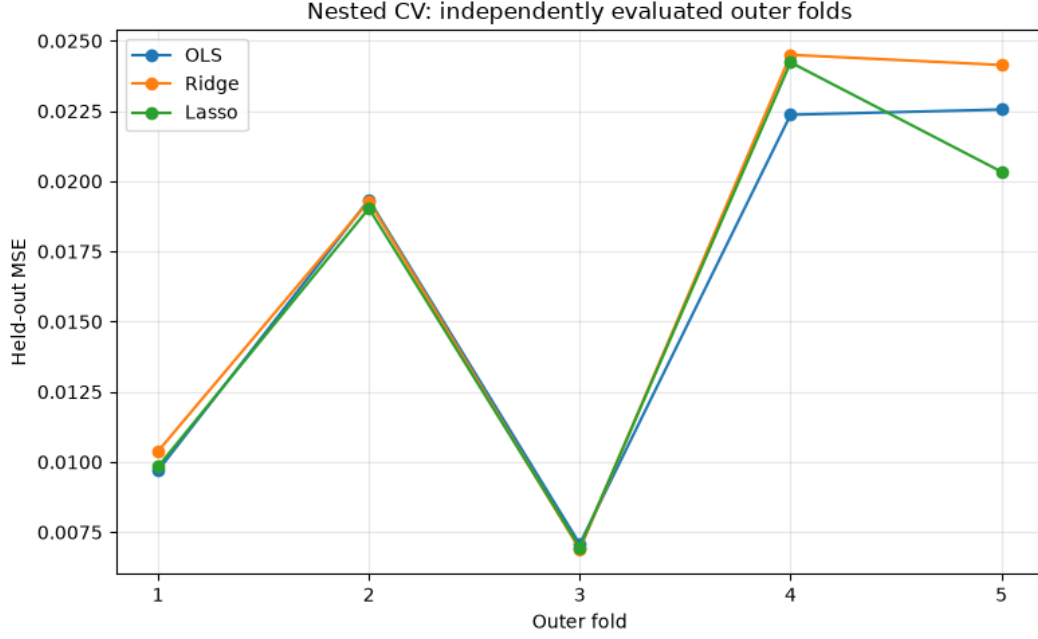


Figure 9: Regenerated held-out MSE for each outer fold. Degree and penalty selection used only the corresponding outer training set's inner folds.

## 6. Discussion and limitations

The results illustrate the central complexity tradeoff. Low-degree polynomials have too little flexibility to represent the Runge function, while very high-degree polynomials fit the training data increasingly well but perform worse on held-out data. The best degree depends on how the error is estimated: the fixed split and cross-validation both selected degree 12 for the full-data OLS/Ridge selection comparisons, while the bootstrap prediction-MSE curve selected degree 4. Nested CV selected different degrees across outer folds, reflecting sensitivity to the available training data. The bootstrap/CV difference is explained in Section 5.3.

Exploratory sample-size curves vary rather than improve monotonically: the lowest fixed-split test MSEs are 0.016193 at  $n = 20$ , 0.005584 at  $n = 50$ , 0.012655 at  $n = 150$ , 0.014924 at  $n = 250$ , and 0.011938 at  $n = 500$ . Each sample size generates a different grid and noise vector, so these single-run minima do not isolate the effect of more observations. For the noise sweep, the lowest exploratory test MSE rises from 0.001090 at  $\sigma = 0$  to 0.137343 at  $\sigma = 0.5$ ; at the latter level the best fixed-split degree is 4 rather than 12 at  $\sigma = 0.01$ , 0.05, and 0.2. Across ten seeds, degree 11 has the lowest *mean* test MSE (0.014270, SD 0.005893), rather than the main fixed split's degree 12. These are sensitivity diagnostics, not separate independent generalization estimates for selected models.

Scaling the input using only training data is appropriate because it avoids using held-out information during preprocessing. It also makes the polynomial design matrix better behaved numerically than using unscaled powers directly. The reported coefficient values therefore belong to the scaled-coordinate polynomial basis.

The Ridge results support the expected interpretation of regularization: small penalties can reduce variance without greatly changing the fit, whereas larger penalties shrink the coefficients more strongly and can increase bias. The full-data selection scores favor mild Ridge regularization slightly, but the nested held-out evaluation favors Lasso and OLS slightly over Ridge. This does not establish that one method is consistently better for other noise realizations or fold partitions.

The optimizer comparison shows that convergence depends strongly on the learning-rate and update rule. The closed-form OLS and Ridge solutions provide useful reference points for testing iterative methods. In particular, the stored results show that Adam reached a small coefficient error much faster than the plain method for the selected settings. This conclusion is specific to the reported learning rates and stopping criteria.

There are also limitations in the numerical evidence. The notebook uses one main 80/20 exploratory split, full-data cross-validation for model selection, and nested cross-validation for independent outer-fold evaluation. There is no additional external test data set, and the same synthetic realization was used in earlier exploratory work. Nested evaluation separates automated selection from outer evaluation, but does not erase the influence of prior exploration on the choice of candidate models. Some experiments are represented by figures without printed numerical tables, so their exact plotted values cannot be reproduced from the text alone. All saved notebook outputs and report figures now correspond to the corrected objective and metric implementations after the fresh-kernel run. Lasso stationarity is checked for this grid, but the comparison remains limited to one synthetic realization and one choice of candidate penalties. The equal-epoch SGD study is not an equal-compute benchmark. Finally, the experiments do not provide a systematic timing benchmark for computational cost, even though the project asks for discussion of computational efficiency.

## 7. Conclusions and future work

I implemented and compared OLS, Ridge, and Lasso regression for noisy observations of the Runge function. OLS and Ridge both selected degree 12 in the full-data cross-validation selection analyses. Ridge with  $\lambda = 10^{-4}$  achieved the lowest selection MSE, 0.01330909, versus 0.01352575 for OLS. These minima are not final evaluation scores. Nested CV gave held-out MSEs of 0.016207633 for OLS, 0.017040405 for Ridge, and 0.016078834 for Lasso. The small differences do not establish a reliable winner. Unlike the proximal-gradient exercise, this Lasso comparison uses a solver whose stationarity was checked on every fit.

The experiments demonstrated that training error alone is not sufficient for selecting polynomial degree. Test errors, bootstrap estimates, and cross-validation reveal the effects of bias and variance. Ridge regularization slightly improved the exploratory and full-data selection scores, but did not improve the pooled nested-CV evaluation over OLS or Lasso in this partition. The iterative methods could approach the closed-form solutions, but their performance depended on the learning-rate schedule and optimizer.

Future work could include a systematic learning-rate and epoch study, timing measurements for the closed-form and iterative methods, repeated nested cross-validation over additional noise realizations, wider Lasso penalty grids, equal-update SGD comparisons, and a more complete numerical comparison of the bootstrap and cross-validation estimates. These would make the conclusions less dependent on a single split and would provide a stronger computational benchmark.

## Appendix: Use of AI/LLM Tools

### Tools used

- ChatGPT, accessed 28 September–4 October 2026 (GPT-5.6 Luna, accessed via <https://chatgpt.com>)
- GitHub Copilot (integrated in IntelliJ), used for subsequent code corrections, nested evaluation, result regeneration, and report updates on 5 October 2026.

## Text writing and editing

The report was drafted from the project notebook and the course documents with ChatGPT assistance. The contribution levels below describe this report-writing interaction. The report was read, understood, reviewed and endorsed by me before submission.

| Report section                                | LLM contribution level | Notes                                                                               |
|-----------------------------------------------|------------------------|-------------------------------------------------------------------------------------|
| Abstract                                      | 3                      | Drafted from the notebook's stored results; numerical claims were kept as printed.  |
| Introduction                                  | 3                      | Drafted from the assignment and solution scope.                                     |
| Theory and formalism                          | 3                      | Drafted from the methods present in the notebook and assignment.                    |
| Data and experimental setup                   | 3                      | Drafted from configuration values and code.                                         |
| Implementation, testing, and numerical checks | 3                      | Drafted from code and saved outputs, including the Ridge normalization discrepancy. |
| Results and analysis                          | 3                      | Drafted from saved printed results and descriptions of stored plots.                |
| Discussion and limitations                    | 3                      | Drafted only from issues visible in the files and stored outputs.                   |
| Conclusions and future work                   | 3                      | Drafted from the reported comparisons and supported limitations.                    |
| AI/LLM appendix                               | 1                      | Spell-check and grammar only.                                                       |
| References                                    | 0                      | No LLM contribution.                                                                |

For the project itself, ChatGPT was used for research and debugging during 28 September–4 October 2026. It did not substantially write or rewrite the submitted code. I verified its assistance by running the code and checking the source material used for the research. The model choices, experiments, numerical results, and final scientific interpretation are my responsibility.

After reviewing the completed notebook, GitHub Copilot acted as a debugging and review aid to identify and correct three groups of issues. GitHub Copilot made the changes in `project1_runge.ipynb`: it made the Ridge penalty convention consistent across the cost, analytical gradient, automatic-differentiation check, and iterative methods; separated held-out bootstrap test MSE from noiseless-grid prediction error and aligned the SGD penalty with the Ridge objective; and connected the gradient-descent routine to its analytical and automatic-differentiation gradient functions. It also added the notebook disclosure required by the course guidelines.

## Code generation and assistance

| File / notebook                          | LLM level | Description                                                                                        |
|------------------------------------------|-----------|----------------------------------------------------------------------------------------------------|
| <code>project1_runge.ipynb</code>        | 2         | Github Copilot used as a debugging and reviewing tool.                                             |
| <code>scripts/run_notebook.py</code>     | 4         | GitHub Copilot generated this script to run the notebook in a fresh kernel and export all figures. |
| <code>scripts/build_report_pdf.py</code> | 4         | GitHub Copilot generated this script to build the report PDF from the Markdown file.               |

## References

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